Navy Genomic Laboratory

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Head of Laboratory: Dr Helen Ashby

Molecular Genetics Report: APC Analysis

Patient name:	**QUINTERO** Rosa	Referred by:	Dr Brennan
		Unit:	Cedar Ridge Genetics Clinic
DOB:	22 Jan 1981	Date requested:	29 Jul 2023
Sex:	Female	Date collected:	30 Jul 2023
NHS No.:	999 040 8185		
Unit no.:	Not provided		
Test reason:	Family history of cancer.		

Results and interpretation:

Test	Result	Lab ref
APC Gene Sequencing and Deletion/Duplication Analysis	**No mutation detected**	NG89-7208

The test sequenced the APC gene and checked it for deletions and duplications. No reportable variant was found. A negative result does not rule out familial adenomatous polyposis 1, because this test can miss some changes and cannot see changes in other genes.

The family history of cancer is still the main reason to plan screening. Colon cancer screening should follow the guidelines for people with a strong family history, and the results of this test should be read together with that history. If a relative has a known APC variant, testing for that same variant is more informative than this test.

Review the family history with a genetic counselor, because a negative result does not end the question of inherited risk. Follow colon cancer screening guidelines for people with a strong family history, which may call for earlier or more frequent colonoscopy than for the general population. If a relative has a known APC variant, ask whether testing for that specific variant is appropriate.

Summary:

No mutation was detected in the gene tested.

Reported: Dr Samuel Grey — *Clinical Scientist*

Authorised: Dr Helen Ashby — *Consultant Clinical Scientist*

Date: 11 Oct 2023

A result is only as reliable as the sample labelling, the diagnoses and the family relationships given on the referral.

Notes: All coding exons of APC and their splice sites were sequenced, and exon copy number was measured by MLPA. Mutations deep within introns or in regulatory regions are not found.

Mutations are named as HGVS recommends, on the reference sequence NM_000038.6.

Rare sequence changes under a primer or probe can give a false positive or a false negative result. For more about this test, see www.navy.example.org.

PROVIDED CLINICAL PHENOTYPE

As written by the ordering clinician on the test requisition form.

- Multiple adenomatous colon polyps
- Odontoma
- Carcinoma
- Congenital hypertrophy of retinal pigment epithelium
- Hyperpigmentation of the skin